

2	(a) sum / total momentum of bodies is constant	B1
	or	
	sum / total momentum of bodies before = sum / total momentum of bodies after	
	for an isolated / closed system / no (resultant) external force	B1
	(b) (i) $4.0 v_A = 6.0 v_B$	C1
	$E_K = \frac{1}{2}mv^2$	C1
	ratio = $\frac{(0.5)(4.0)(6.0)^2}{(0.5)(6.0)(4.0)^2}$	M1
	= 1.5	A0
	(ii) $0.48 = E_K \text{ of A} + E_K \text{ of B}$	
	= $E_K \text{ of A} + (E_K \text{ of A} / 1.5) = 5/3 \times E_K \text{ of A}$	C1
	$E_K \text{ of A} = 0.29 \text{ (0.288) J}$	A1
	(iii) curve starts from origin and has decreasing gradient	M1
	final gradient of graph line is zero	A1

3 (a) (i) $E_K = \frac{1}{2}mv^2$